VISHNUVARDHAN REDDY KASTHURI

Frisco, TX | 513-996-8470 | vishnuvardhanrk30@gmail.com

SUMMARY

- **Data Engineer** with over **3+ Years** of hands-on experience in developing and implementing high- performance data solutions.
- Efficient in preprocessing data including Data cleaning, Correlation analysis, Imputation, Visualization, Feature Scaling and Dimensionality Reduction techniques using Machine learning platforms like Python Data Science Packages (Scikit-Learn, Pandas, NumPy).
- Proficient in programming languages such as Python and SQL and experienced with large-scale data processing technologies like Apache Spark.
- Skilled in leveraging cloud platforms, particularly AWS and Azure, for architecting ETL processes, managing data repositories, and constructing data integration workflows.
- Proficient in developing scalable and maintainable Java applications using Spring Boot and Hibernate frameworks.
- Proficient in creating and optimizing CI/CD workflows and adhering to DataOps best practices.
- Familiar with key data science concepts (statistics, data visualization, machine learning, etc.). Experienced in Python, R, MATLAB, SAS, PySpark programming for statistic and quantitative analysis.
- Experience working with Big Data tools such as Hadoop - HDFS and MapReduce, Hive QL, Sqoop, Pig Latin and Apache Spark (PySpark).
- Knowledge and experience working in Agile environments including the scrum process and used Project Management tools like ProjectLibre, Jira and version control tools such as GitHub/Git.

SKILLS

- **Programming Languages:** Python, Scala, SQL, Java, C++, Go
- **Big Data Technologies:** Apache Spark (Spark SQL, PySpark, Spark Streaming), Hadoop, Hive, Impala, Databricks (Databricks SQL, Delta Lake, REST API)
- **Cloud Platforms:** AWS (EMR, Redshift, S3, Lambda, Glue, CloudFormation), Azure (Azure Data Factory, Azure Synapse Analytics, Azure DevOps), Google Cloud Platform (GCP) (BigQuery, Dataflow, Dataproc)
- **ETL Tools and Data Pipeline Software:** Airflow, Azure Data Factory, Prefect, DBT, Apache NiFi, Autosys
- **Data Warehousing:** Snowflake, PostgreSQL, MySQL, Redshift
- **Data Modeling and Data Management:** Data modeling (relational and NoSQL), Data governance (e.g., Alation), Data quality, DataOps tools, Data profiling, Data transformation, SSIS.
- **Containerization and Orchestration:** Kubernetes (K8s), Docker, Terraform
- **Version Control and CI/CD:** GitHub (GitHub Actions), Jenkins, GitLab CI
- **Visualization and Reporting Tools:** Tableau, Power BI
- **Additional Technologies:** Apache Kafka, Pandas, UNIX Shell scripting, Ansible, Cloudera suite (Hive, Impala, Spark), ElastiCache, DynamoDB
- **AWS Services:** IAM roles, AWS Glue, AWS Lambda, AWS QuickSight

PROFESSIONAL EXPERIENCE

Verizon Wireless Systems — **Jan 2024 - Current**

Data Engineer

- Engineered and implemented robust data integration solutions using Databricks and AWS Glue, efficiently managing over 5TB of data per day.
- Worked on Amazon Web Services (AWS) cloud services to do machine learning on big data.
- Enhanced ETL processes by utilizing Spark and Delta Lake, achieving a 40% reduction in processing times.
- Partnered with data scientists to embed machine learning models into production data workflows, boosting predictive analytics capabilities.
- Administered AWS services including EC2, VPC, S3, ELB, Glacier, Route 53, IAM, and Trusted Advisor, ensuring efficient data storage and processing.
- Designed and executed data migration pipelines to transfer data from MySQL and PostgreSQL using PySpark, guaranteeing data integrity and accuracy throughout the process.
- Developed and optimized data processing workflows using PySpark on AWS, leveraging its distributed computing capabilities to efficiently handle and analyze large-scale datasets.
- Managed the ingestion and processing of structured and semi-structured data on AWS using S3 and Python, facilitating the migration of on-premises big data workloads to AWS.
- Developed code for feature extraction, applied Principal Component Analysis (PCA), and fine-tuned hyper parameters to improve model performance.
- Utilized Terraform for automating infrastructure provisioning on AWS, ensuring consistent and repeatable deployments ofdata solutions.
- Designed and implemented database solutions utilizing SSIS for data integration, including the creation and testing of stored procedures, views, and triggers to streamline data workflows and ensure system efficienc
- Explored ensemble techniques to enhance model accuracy through different bagging and boosting strategies and deployed the model on AWS.
- Created a Python-based RESTful API for revenue tracking and analytics, streamlining data access and reporting processes.

- Used MySQL Workbench and Birst to optimize database performance, reducing query execution times by 25% and speeding up report generation by 20%. Utilized apache airflow for data orchestration workflows.

Coforge **Jan 2020 - July 2022**

Data Engineer

- Engaged in big data requirement analysis to develop and design ETL and Business Intelligence solutions, ensuring alignment with business needs.
- Performed data cleaning including transforming variables and dealing with missing value and ensured data quality, consistency, integrity using Pandas, NumPy.
- Developed various machine learning models such as Logistic regression, KNN, and Gradient Boosting with Pandas, NumPy, Seaborn, Matplotlib, Scikit-learn in Python.
- Designed and implemented robust ETL pipelines on AWS using AWS Glue and Amazon Redshift, ensuring high-performance data integration and analytics for enterprise applications.
- Developed Spark Python modules for machine learning & predictive analytics in Hadoop.
- Created 3NF data models for Operational Data Stores (ODS) and Online Transaction Processing (OLTP) systems, as well as dimensional data models using Star and Snowflake Schemas.
- Worked within the Snowflake environment to eliminate data redundancy and integrate real-time data from various sources into HDFS via Kafka. Developed a comprehensive data warehouse model in Snowflake, incorporating over 100datasets.
- Engineered and managed data pipelines on AWS using Apache Spark and AWS Glue, enabling theseamless ingestion, transformation, and analysis of large volumes of data.
- Designed data aggregation processes in Hive for ETL tasks on Amazon EMR, ensuring that processes met business requirements.
- Implemented data validation through MapReduce programs to filter out unnecessary records before transferring data into Hive tables.
- Extracted real-time data feeds using Kafka and Spark Streaming, converting them into RDDs, processing them as Data Frames, and saving the results in Parquet format within HDFS.

EDUCATION

- **Master of Science, Information Technology**
  University of Cincinnati, Cincinnati, Ohio
- **Bachelor of Technology, Electronics and Communication Engineering.**
  CVR College of Engineering, Hyderabad, India

CERTIFICATIONS

- **AWS Associate Data Engineer**
- **Microsoft Azure: AZ-900 Fundamentals**
- **Google IT Professional Course**
- **Python Essential Learning, LinkedIn**